Week 12

Advanced Link Analysis

Module 2 of Phase-1 Final Prep · Builds directly on W8-9 PageRank

Topics covered in this PDF:

- Three limitations of vanilla PageRank
- Topic-Sensitive PageRank — biased teleport set
- Web spam: term spam & Google's anchor-text fix
- Link spam & spam farms — analysis with multiplier formula
- TrustRank — topic-sensitive PR with trusted seeds
- Spam Mass estimation — fractional spam contribution
- HITS algorithm — Hubs & Authorities (brief)
- 6 worked numerical examples · 15 practice questions

CS-404 Big Data Analytics · Dr. Syed Imran Ali · Final ~ 2026-05-16

Prepared as a comprehensive professor-grade study tutor.

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§ 1 BEGINNING KEY NOTES — STUDY COMPASS

THE 9 IDEAS YOU MUST OWN

These build directly on the W8-9 PageRank machinery — make sure you can recite $r = \beta \cdot M \cdot r + (1-\beta)/N \cdot 1_N$ from memory before starting this PDF.

• **Three weaknesses of vanilla PageRank:** (1) it gives generic popularity, missing topic-specific authorities; (2) it uses a single notion of importance — HITS provides hubs vs. authorities; (3) it is gameable through link spam — TrustRank fixes this.

• **Topic-Sensitive PageRank:** change the teleport target. Instead of jumping uniformly to ANY page, jump uniformly to a small *topic set S*. One PageRank vector per topic.

• **Topic-PR formula:** $A_{ij} = \beta \cdot M_{ij} + (1-\beta)/|S|$ if $j \in S$, else $\beta \cdot M_{ij}$. Equivalently: $r = \beta \cdot M \cdot r + (1-\beta) \cdot v_S$, where v_S has $1/|S|$ in positions S and 0 elsewhere.

• **Term spam:** stuffing pages with hidden keywords. Google's defense: trust anchor-text from incoming links, not page's own words.

• **Link spam & spam farms:** three page types — inaccessible / accessible / owned. By creating M owned pages all linking to target t and sneaking links from a small accessible set x, spammer multiplies $y = x/(1-\beta^2) + c \cdot M/N$ where $c = \beta/(1+\beta)$.

• **Multiplier value:** for $\beta=0.85$, $1/(1-\beta^2) \approx 3.6$ — a small accessible PR x gets amplified ~3.6×.

• **TrustRank:** topic-sensitive PageRank with the teleport set = trusted human-vetted seed pages (typically .edu, .gov, .mil domains).

• **Trust attenuation & splitting:** trust decays with graph distance and is divided across out-links — captured by the same algebra as PageRank, with seed teleport.

• **Spam Mass:** $r^-_p = r_p - r^+_p$, spam mass of p = r^-_p / r_p. Pages with mass close to 1 are spam.

• **HITS:** two scores — h_i (hub) and a_j (authority) — found by iterating $h \leftarrow M \cdot a$ and $a \leftarrow M^T \cdot h$ with normalization. Different from PageRank; both are exam-bait concepts.

§ 2 THREE LIMITATIONS OF GENERIC PAGERANK

Vanilla PageRank gives a single rank per page reflecting overall web-graph importance. This works well for queries like “Stanford” but fails for three reasons:

(1) No topic awareness. A query for the word “Trojan” legitimately maps to three different topics — sports (USC Trojans), history (Trojan war), or computer security (Trojan malware). Generic PageRank cannot prefer the right interpretation given the user's actual interest.

(2) Single notion of importance. Some pages are *good answer pages* (authoritative content), and some are *good index pages* (useful pointers). PageRank conflates these. The HITS algorithm splits importance into two scores: **hubs** (pages that point to many authorities) and **authorities** (pages pointed to by many hubs).

(3) Spam vulnerability. The algorithm is purely structural — it cannot tell whether an incoming link is genuine or spam-farmed. The fix is TrustRank (§6).

§ 3 TOPIC-SENSITIVE PAGERANK

Idea. Recall the random surfer with teleport probability 1−β. Vanilla PageRank teleports to a uniformly random page (every page equally likely). **Topic-sensitive PageRank** instead teleports to a uniformly random page from a *topic-relevant set* S — pages known to be about a particular topic, for example all DMOZ-classified “sports” pages.

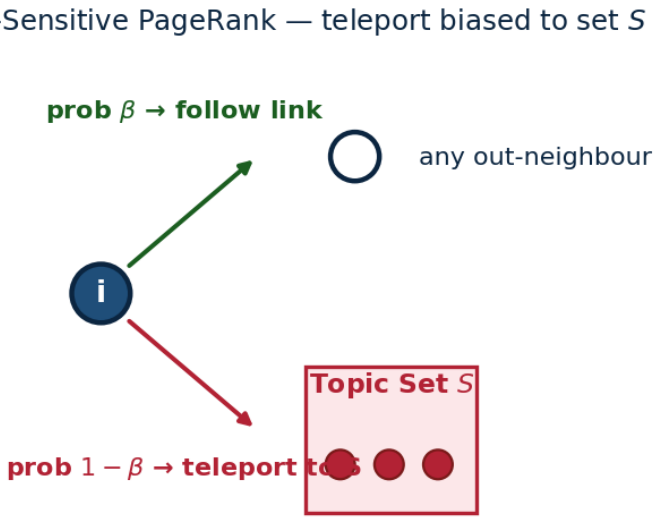

Figure 3.1. *Topic-sensitive teleport. With probability β the surfer follows a link as before; with probability 1−β the surfer jumps uniformly into the topic set S, never to pages outside S.*

Matrix formulation. Define the topic-sensitive Google matrix:

$$A_{ij} = \beta M_{ij} + \frac{1-\beta}{|S|} \text{ if } j \in S, \quad A_{ij} = \beta M_{ij} \text{ otherwise}$$

Equivalently — and this is the form most often used on exams:

$$r = \beta \cdot M \cdot r + (1-\beta) \cdot v_S$$

where $\mathbf{v_S}$ is a vector with 1/|S| in positions corresponding to pages in S, and 0 elsewhere. Plugging |S| = N and $v_S = (1/N) \cdot 1_N$ recovers vanilla PageRank — confirming Topic-PR is a strict generalization.

One vector per topic. Compute and store r_{S_1}, r_{S_2}, … for the 16 DMOZ top-level categories (arts, business, sports, …). At query time, pick the right vector based on inferred topic.

EXAM-STYLE ALGORITHM CHANGE — only one line

The complete algorithm from W8-9 changes by exactly one line: replace the *uniform redistribution* step c. with: $r^{new}_j = r'^{new}_j + (1-S)/|S|$ if $j \in S$, else r'^{new}_j. Everything else (initialization, $\beta \cdot M \cdot r^{old}$, leak handling) stays identical.

§ 4 WEB SPAM — TERM SPAM AND GOOGLE'S DEFENSE

Spamming is any deliberate action to boost a page's search-engine ranking beyond its real value. **Spam** is the resulting page. Approximately 10–15% of web pages are spam — huge enough that pretending it doesn't exist breaks the search engine.

First-generation spam (term spam). Early search engines ranked by word-frequency on the page. Spammers exploited this in two ways:

- **Keyword stuffing.** Add the target keyword (e.g. “movie”) thousands of times, with text color matching the background so users don't see it but the crawler does.
- **Top-result mimicry.** Run the query “movie” yourself, see which page comes first, copy its content invisibly into your own page.

Google's term-spam fix: believe what others say about you, not what you say about yourself. Specifically, weight the **anchor text** of incoming links — the visible text inside <a> tags pointing TO your page — much more than your page's own body text. Combined with PageRank, this surfaced authentic, well-cited pages while burying term-spam farms.

WHY ANCHOR-TEXT WORKS

If 1000 pages link to a movie review with anchor text “great movie”, that's 1000 independent judgments. The reviewed page can't game this without compromising the linking pages — which themselves have to convince their own users to keep visiting. Distributed truth-telling.

§ 5 LINK SPAM & SPAM FARMS

Once Google adopted PageRank + anchor text as the ranking signal, spammers shifted to **link spam**: artificially constructing link structures to boost the PageRank of a target page t.

Three page types from the spammer's view:

- **Inaccessible** pages — controlled by others, no influence possible.
- **Accessible** pages — e.g. blog comment sections where the spammer can post a link.
- **Owned** pages — completely under the spammer's control, possibly across many domains.

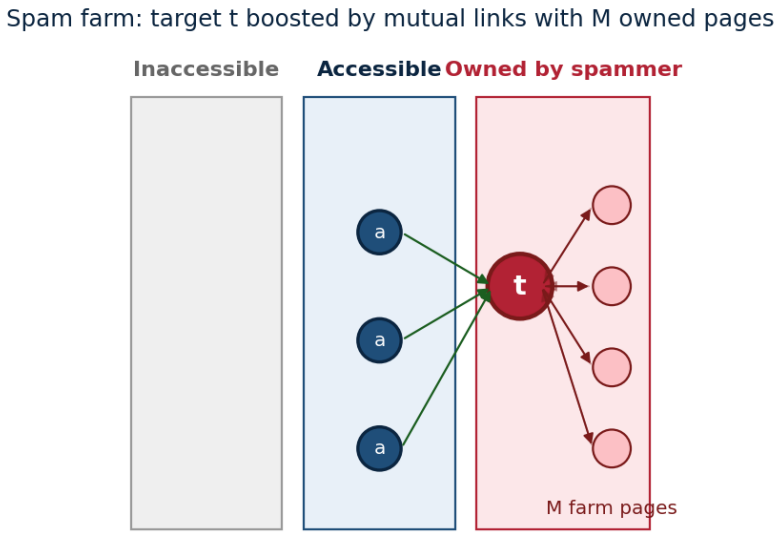

Figure 5.1. *A canonical spam farm. The spammer scatters links from accessible pages (e.g. blog comments) to the target t, then links t to and from M owned "farm" pages, all of which link back to t — concentrating PageRank on t.*

Spam-farm analysis. Let:

- **x** = total PageRank contributed to t by accessible pages (small, but nonzero).
- **y** = PageRank of target t (what we want to estimate).
- **M** = number of farm pages owned by spammer; **N** = total pages on the web.

Each farm page receives β·y/M from t (t splits its rank among its M owned out-links) plus the teleport (1−β)/N. So:

$$\text{rank}_{\text{farm}} = \beta \cdot \frac{y}{M} + \frac{1-\beta}{N}$$

Then t's rank is x (from accessible) + β·M·$\text{rank}_{\text{farm}}$ (from M farm pages) + (1−β)/N (teleport). Substituting and ignoring the small (1−β)/N term:

$$y = x + \beta M \cdot \left(\frac{\beta y}{M} + \frac{1-\beta}{N}\right) \Rightarrow y = x + \beta^2 y + \beta(1-\beta)\frac{M}{N}$$

Solving for y:

$$y = \frac{x}{1-\beta^2} + c \cdot \frac{M}{N}, \quad c = \frac{\beta}{1+\beta}$$

Multiplier. For β = 0.85, the multiplier 1/(1−β²) ≈ 3.6. Even a small accessible-PR contribution x of 0.01 becomes y ≈ 0.036 + c·M/N. By making M (the farm size) huge, the spammer can push y arbitrarily high — that is the spam farm's "multiplier effect".

EXAM TRAP — careful with β vs β² in the denominator

The denominator is 1 − β², not 1 − β. Many students write 1 − β by mistake. The β² comes from the round-trip rank flow: t → farm → t. If you skip the algebra, remember: *two β's, two M's cancel* → β² survives.

§ 6 TRUSTRANK — COMBATING LINK SPAM

Core insight (approximate isolation). Trustworthy pages tend not to link to spam pages. If we can identify a small set of **trusted seed pages** by hand, we can propagate

their trust outward through the link graph — pages reachable from trusted seeds in few hops are likely good; pages far from any trusted seed are likely spam.

Algorithm. TrustRank is exactly Topic-Sensitive PageRank with the topic set S = trusted seed pages. Each page receives a trust value $t_p \in [0, 1]$.

- Set $t_p = 1$ for every trusted seed page, 0 elsewhere initially.
- Iterate: each page p with out-link set O_p contributes $\beta \cdot t_p / |O_p|$ to each $q \in O_p$.
- Sum incoming contributions per page; converge as in PageRank.

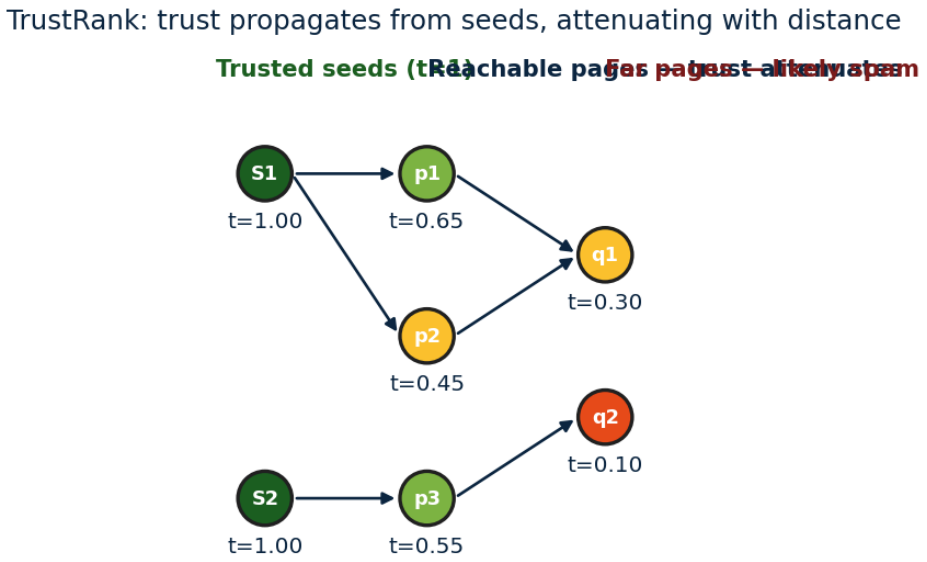

Figure 6.1. Trust propagation. Seed pages S1, S2 have t = 1.0. First-hop pages inherit ~0.5–0.65 trust, second-hop pages inherit less. Spam pages disconnected from any seed never accumulate significant trust.

Two effects you should be able to articulate on the exam:

- **Trust attenuation** — trust decreases with graph distance from seeds, because at every hop only a β fraction is propagated and the rest leaks into teleport.
- **Trust splitting** — a page with many out-links splits its trust thinly across them. Captures the intuition that an editor with 1000 links per page reviews each less carefully.

Picking the seed set. Two competing goals — small enough that humans can vet every page, yet broad enough that every legitimate web region is reachable in few hops. Two practical tactics:

- Take the top-k pages by vanilla PageRank (theory: spam can't reach the very top).
- Restrict to controlled-membership domains: *.edu*, *.gov*, *.mil*.

§ 7 SPAM MASS — ESTIMATING FRACTIONAL SPAM CONTRIBUTION

Complementary view. TrustRank tells us how much of a page's rank comes from *trusted* sources. Spam Mass asks the converse: how much of a page's rank comes from *spam*?

Let:

- r_p = vanilla PageRank of page p.
- r^+_p = TrustRank of p (= topic-sensitive PR with teleport into trusted seed set).

- $r^-_p = r_p - r^+_p$ = rank contributed by non-trusted (likely-spam) sources.

Spam Mass of page p:

$$\text{SpamMass}(p) = \frac{r_p^-}{r_p} = \frac{r_p - r_p^+}{r_p}$$

Spam Mass close to 1 means almost all of p's rank comes from non-trusted (spam) sources → p is likely spam. Spam Mass close to 0 means most of p's rank comes from trusted sources → p is legitimate. Pick a threshold (e.g. 0.5) and label everything above it as spam.

WHY USE BOTH r AND r^+?

If we only had TrustRank, we'd label every page far from the seed set as spam — including perfectly legitimate isolated communities (small academic groups, niche hobbyist sites). Spam Mass instead asks: *relative to its overall rank, how much of this page's rank is attributable to spam-like sources?* A small but legitimate page has small r and small r^+, so SpamMass is small. Only pages with high r and proportionally low r^+ get flagged.

§ 8 HITS — HUBS & AUTHORITIES (BRIEF)

HITS (Hyperlink-Induced Topic Search) is the second classical link-analysis algorithm. The slides only mention HITS in passing as *"Solution: Hubs-and-Authorities"* — but the concept is fair game on conceptual exam questions because it's in MMDS §5.5.

Two scores per page, jointly defined:

- **Hub score h_i** — high if i links TO many high-authority pages.
- **Authority score a_j** — high if j is linked FROM many high-hub pages.

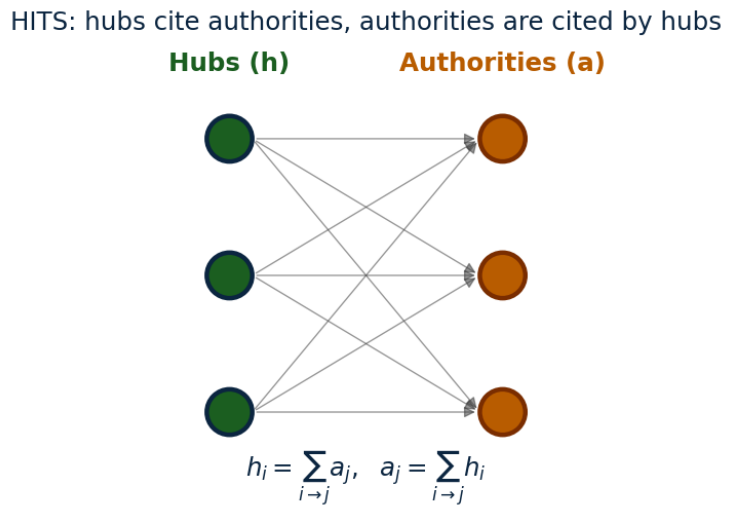

Figure 8.1. *HITS bipartite intuition. Each hub points to multiple authorities; authorities receive links from multiple hubs. The two scores reinforce each other.*

Iteration. Let A be the link adjacency matrix (A_{ij}=1 if i→j). Then:

$$h = A \cdot a, \quad a = A^T \cdot h$$

Substituting one into the other: $a = A^T \cdot A \cdot a$, $h = A \cdot A^T \cdot h$ — so a is the principal eigenvector of $A^T \cdot A$ and h is the principal eigenvector of $A \cdot A^T$. Power-iterate, normalizing each step (typically L1 or L2).

PAGERANK vs HITS — common compare-and-contrast question

- **PageRank**: one score per page, computed offline (query-independent), uses teleport.
- **HITS**: two scores per page, traditionally computed on a query-specific subgraph (root + base set), no teleport.
- PageRank uses column-stochastic M; HITS uses raw adjacency A.
- PageRank converges from any positive start due to teleport; HITS needs explicit normalization to converge.
- Both are eigenvector-based; both reduce to power iteration.

§ 9 SIX WORKED NUMERICAL EXAMPLES

Worked Example 1 — Topic-Sensitive PageRank, one iteration

Problem. The 4-node graph from the W12 slides has edges: 1→2, 1→3, 2→3, 3→1, 3→4, 4→1, 4→2 (out-degrees: d_1=2, d_2=1, d_3=2, d_4=2). Topic set S = {1}, β = 0.8. Compute one iteration starting from $r^{(0)}$ = (1/4, 1/4, 1/4, 1/4).

Step 1 — M (column-stochastic).

	fr 1	fr 2	fr 3	fr 4
row 1	0	0	1/2	1/2
row 2	1/2	0	0	1/2
row 3	1/2	1	0	0
row 4	0	0	1/2	0

Step 2 — M·$r^{(0)}$. Each entry is column·(1/4):

- r'_1 = 0+0+(1/2)(1/4)+(1/2)(1/4) = 0.250
- r'_2 = (1/2)(1/4)+0+0+(1/2)(1/4) = 0.250
- r'_3 = (1/2)(1/4)+1·(1/4)+0+0 = 0.375
- r'_4 = 0+0+(1/2)(1/4)+0 = 0.125

Step 3 — β·M·$r^{(0)}$. Multiply each by 0.8: (0.200, 0.200, 0.300, 0.100). Sum S = 0.800.

Step 4 — add teleport into S = {1} only. (1 − β)/|S| = 0.2/1 = 0.2 added ONLY to node 1. Other nodes get 0.

→ $r^{(1)}$ = (0.200+0.200, 0.200, 0.300, 0.100) = **(0.400, 0.200, 0.300, 0.100)**. Sum = 1.000 ✓.

Note how node 1 jumps from 0.25 to 0.40 — the teleport mass is now concentrated there. Slide answer (after convergence): r ≈ (0.29, 0.11, 0.32, 0.26). Direction matches.

Worked Example 2 — Topic set with two pages

Problem. Same 4-node graph as WE 1, but now S = {1, 2} and β = 0.8. Compute one iteration starting from $r^{(0)}$ uniform.

β·M·$r^{(0)}$ = (0.200, 0.200, 0.300, 0.100) — same as WE 1.

Teleport: (1−β)/|S| = 0.2/2 = 0.1 added to nodes 1 AND 2; others get 0.

→ $r^{(1)}$ = (0.200+0.1, 0.200+0.1, 0.300, 0.100) = **(0.300, 0.300, 0.300, 0.100)**. Sum = 1.000 ✓.

Spreading the teleport across two seeds reduces the boost to each but covers more topical scope.

Worked Example 3 — Spam-farm multiplier in numbers

Problem. Spammer creates a farm with $M = 10^5$ owned pages all linking to target t. External (accessible) PR contribution to t is $x = 10^{-4}$. Total web $N = 10^9$ pages. $\beta = 0.85$. Compute the spam farm's PageRank y.

Apply formula: $y = x/(1-\beta^2) + c \cdot M/N$ where $c = \beta/(1+\beta)$.

- $1 - \beta^2 = 1 - 0.7225 = 0.2775$. → $x/(1-\beta^2) = 10^{-4}/0.2775 \approx$ **3.604×10^{-4}**.
- $c = 0.85/1.85 \approx 0.459$. → $c \cdot M/N = 0.459 \cdot 10^5/10^9 = 4.59 \times 10^{-5}$.
- $y \approx 3.604 \times 10^{-4} + 4.59 \times 10^{-5} \approx$ **4.06×10^{-4}**.

Sanity check. Without the farm, t's rank would be $\approx x = 10^{-4}$. The farm amplifies it by ~4×. With $M = 10^6$ owned pages, the $c \cdot M/N$ term grows to 4.59×10^{-4} — the farm's structural contribution dominates. That's the “multiplier effect” in action.

Worked Example 4 — TrustRank, one iteration

Problem. 5 pages: A, B, C, D, E. Edges: A→B, B→A, B→C, C→D, D→E, E→D. Trusted seed set: {A}. $\beta = 0.85$. Compute one iteration of TrustRank.

Out-degrees: $d_A=1$, $d_B=2$, $d_C=1$, $d_D=1$, $d_E=1$.

Initial $r^{(0)}$: $= (1/5) \cdot 1 = 0.2$ each (TrustRank starts uniform like PageRank).

$M \cdot r^{(0)}$:

- $r'_A = (1/2) \cdot 0.2 = 0.100$ (only B→A)
- $r'_B = 1 \cdot 0.2 = 0.200$ (only A→B)
- $r'_C = (1/2) \cdot 0.2 = 0.100$ (only B→C)
- $r'_D = 1 \cdot 0.2 + 1 \cdot 0.2 = 0.400$ (C→D and E→D)
- $r'_E = 1 \cdot 0.2 = 0.200$ (D→E)

$\beta \cdot M \cdot r^{(0)}$: $0.85 \cdot (0.100, 0.200, 0.100, 0.400, 0.200) = (0.085, 0.170, 0.085, 0.340, 0.170)$. Sum S = 0.850.

Add teleport into seed set {A}: $(1-\beta)/|S| = 0.15/1 = 0.15$ to A only.

→ $r^{(1)}$ = **(0.235, 0.170, 0.085, 0.340, 0.170)**. Sum = 1.000 ✓.

Notice A's rank jumps because it's the only teleport target. After more iterations, B (close to A) accumulates more trust than E (far from A) — exactly the trust attenuation effect from §6.

Worked Example 5 — Spam Mass calculation

Problem. A page p has vanilla PageRank $r_p = 0.012$ and TrustRank $r^+_p = 0.003$. (a) Compute Spam Mass. (b) If the threshold is 0.5, is p flagged as spam?

(a) $r^-_p = r_p - r^+_p = 0.012 - 0.003 = 0.009$.

Spam Mass = $r^-_p / r_p = 0.009 / 0.012 =$ **0.75**.

(b) 0.75 > 0.5 → **YES, p is flagged as spam**. ¾ of its PageRank comes from non-trusted sources.

Worked Example 6 — HITS, two iterations

Problem. Three pages with link adjacency: 1→2, 1→3, 2→3, 3→1. Compute hubs h and authorities a for two iterations starting from $h^{(0)}=a^{(0)}=(1,1,1)$. Normalize by L1 norm after each step.

A (raw adjacency): rows = source, cols = target. A = [[0,1,1],[0,0,1],[1,0,0]]. A^T = [[0,0,1],[1,0,0],[1,1,0]].

Iter 1 authorities: $a = A^T \cdot h = (h_3, h_1, h_1+h_2) = (1, 1, 2)$. Normalize: a = (1/4, 1/4, 2/4) = (0.25, 0.25, 0.5).

Iter 1 hubs: $h = A \cdot a = (a_2+a_3, a_3, a_1) = (0.75, 0.5, 0.25)$. Normalize (sum=1.5): h = (0.5, 0.333, 0.167).

Iter 2 authorities: $a = A^T \cdot h = (h_3, h_1, h_1+h_2) = (0.167, 0.5, 0.833)$. Normalize (sum=1.5): a = (0.111, 0.333, 0.556).

Iter 2 hubs: $h = A \cdot a = (a_2+a_3, a_3, a_1) = (0.889, 0.556, 0.111)$. Normalize (sum=1.556): h = (0.571, 0.357, 0.071).

Reading the converged scores: page 3 is the strongest authority (linked from 1 and 2). Page 1 is the strongest hub (links to 2 and 3). Page 2 is weaker on both metrics. These rankings match the link-graph intuition.

§ 10 PRACTICE QUESTIONS — 15

Q1 [Numerical · 6 marks]

For the 4-node graph from WE 1 (edges 1→2, 1→3, 2→3, 3→1, 3→4, 4→1, 4→2), compute Topic-Sensitive PageRank with S = {3} and β = 0.85 for one iteration starting from uniform $r^{(0)}$. Show $M \cdot r^{(0)}$, $\beta \cdot M \cdot r^{(0)}$, the teleport contribution, and the final $r^{(1)}$.

Q2 [Numerical · 5 marks]

Using the same 4-node graph and S = {1, 2, 3, 4} (i.e. teleporting uniformly to all nodes — recovering vanilla PageRank), compute $r^{(1)}$ from uniform $r^{(0)}$ with β = 0.8. Verify the teleport contribution equals $(1-\beta)/N$ for each node.

Q3 [Numerical · 6 marks]

A spammer builds a farm with $M = 5\times10^4$ owned pages, each linking back to target t. External (accessible) PR contribution to t is $x = 5\times10^{-5}$. Total web pages $N = 10^9$. (a) For β = 0.85, compute the multiplier $1/(1-\beta^2)$. (b) Compute y, the PageRank of t. (c) What value of M is needed to make $y \geq 10^{-3}$?

Q4 [Numerical · 7 marks]

Five-page graph: A→B, A→C, B→C, B→D, C→A, D→E, E→C. Trusted seed set = {A, C}. β = 0.8. Compute one iteration of TrustRank from $r^{(0)}$ uniform. List $M \cdot r^{(0)}$ entry-by-entry, compute $\beta \cdot M \cdot r^{(0)}$, then add teleport into the seed set.

Q5 [Numerical · 4 marks]

Page p has vanilla PageRank $r_p = 0.025$ and TrustRank $r^+_p = 0.020$. (a) Compute Spam Mass. (b) If threshold = 0.4, is p flagged? (c) Suppose we re-run TrustRank with a larger seed set and r^+_p rises to 0.022. Does Spam Mass go up or down? By how much?

Q6 [Numerical · 6 marks]

Three-page graph: 1→2, 1→3, 2→1, 2→3, 3→2. Compute HITS hub and authority scores for two iterations starting from $h^{(0)}=a^{(0)}=(1,1,1)$. Normalize by L1 norm at each step. State which page is the strongest authority and which is the strongest hub.

Q7 [Concept · 4 marks]

Explain how Topic-Sensitive PageRank is implemented as a one-line modification of the vanilla PageRank algorithm. Then give a concrete example where Topic-Sensitive PR returns a strictly different ranking from vanilla PR (you may invent a small graph).

Q8 [Concept · 4 marks]

Define **trust attenuation** and **trust splitting**. For each, explain the intuition (one sentence) and identify the corresponding term in the algebra of the TrustRank iteration.

Q9 [Concept · 4 marks]

Walk through the spam-farm derivation. Explain why the round-trip rank flow contributes a β^2 factor and why the $(1-\beta)/N$ teleport term is typically dropped from the final formula. State the multiplier value for β = 0.9.

Q10 [Concept · 4 marks]

Compare PageRank, TrustRank, HITS, and Spam Mass on three axes: (a) what they assign per page, (b) what their teleport / normalization is, (c) what kind of question they answer.

Q11 [Concept · 3 marks]

Why is Spam Mass preferable to thresholding TrustRank directly? Give a concrete scenario where direct TrustRank thresholding misclassifies a legitimate page as spam, but Spam Mass does not.

Q12 [MCQ · 2 marks]

In Topic-Sensitive PageRank with topic set S and |S| pages, the teleport contribution to page $j \in S$ each iteration is: (A) $(1-\beta)/N$, (B) $(1-\beta)/|S|$, (C) $\beta/|S|$, (D) $\beta\cdot(1-\beta)/N$.

Q13 [MCQ · 2 marks]

The spam-farm multiplier for $\beta = 0.5$ is closest to: (A) 1.0, (B) 1.33, (C) 1.6, (D) 2.0.

Q14 [True/False · 2 marks]

True or false: HITS converges from any positive initial vector without explicit normalization. Justify.

Q15 [MCQ · 2 marks]

Which of the following is the MOST robust strategy for picking a TrustRank seed set? (A) random sample of 1000 pages, (B) top-k pages by vanilla PageRank, (C) every page on the web — most are good, (D) only pages from .edu / .gov / .mil domains.

§ 11 FULL WORKED ANSWERS

A1

$M \cdot r^{(0)}$ = (0.250, 0.250, 0.375, 0.125) (from WE 1). $\beta \cdot M \cdot r^{(0)}$ = 0.85 · (0.250, 0.250, 0.375, 0.125) = (0.2125, 0.2125, 0.31875, 0.10625). Sum S = 0.85. $(1-\beta)/|S|$ = 0.15/1 = 0.15 added to node 3 only. → $r^{(1)}$ = (0.2125, 0.2125, 0.46875, 0.10625). Sum ≈ 1.000 ✓. Node 3 surges from 0.31875 to 0.46875 because it's the lone topic seed.

A2

$\beta \cdot M \cdot r^{(0)}$ with β=0.8: 0.8 · (0.250, 0.250, 0.375, 0.125) = (0.200, 0.200, 0.300, 0.100). Sum S = 0.800. Teleport $(1-\beta)/|S|$ with |S|=N=4: 0.2/4 = 0.05 added to every node. → $r^{(1)}$ = (0.250, 0.250, 0.350, 0.150). Sum = 1.000 ✓. Verify: each node gets exactly 0.05 from teleport — equal to $(1-\beta)/N$ = 0.2/4 ✓.

A3

(a) $1/(1-\beta^2) = 1/(1-0.7225) = 1/0.2775 =$ **3.604**.
(b) $c = \beta/(1+\beta) = 0.85/1.85 = 0.4595$.
$y = x/(1-\beta^2) + c \cdot M/N = 5\times10^{-5} \cdot 3.604 + 0.4595 \cdot 5\times10^4/10^9 = 1.802\times10^{-4} + 2.298\times10^{-5} \approx$ $\mathbf{2.03\times10^{-4}}$.
(c) Want $y \geq 10^{-3}$. The first term is fixed at 1.802×10^{-4}. Need $c \cdot M/N \geq 10^{-3} - 1.802\times10^{-4} = 8.198\times10^{-4}$. Solve: $M \geq N \cdot 8.198\times10^{-4}/c = 10^9 \cdot 8.198\times10^{-4}/0.4595 \approx$ $\mathbf{1.78\times10^6}$ pages. So the spammer needs ~1.8 million owned pages to push y above 10^{-3}.

A4

Out-degrees: d_A=2 (A→B, A→C), d_B=2 (B→C, B→D), d_C=1 (C→A), d_D=1 (D→E), d_E=1 (E→C). $r^{(0)}$=(0.2, 0.2, 0.2, 0.2, 0.2).

$M \cdot r^{(0)}$:
- $r'_A = 1 \cdot 0.2 = 0.200$ (only C→A)
- $r'_B = (1/2) \cdot 0.2 = 0.100$ (A→B)
- $r'_C = (1/2) \cdot 0.2 + (1/2) \cdot 0.2 + 1 \cdot 0.2 = 0.500$ (A→C, B→C, E→C)
- $r'_D = (1/2) \cdot 0.2 = 0.100$ (B→D)
- $r'_E = 1 \cdot 0.2 = 0.200$ (D→E)

$\beta \cdot M \cdot r^{(0)}$: 0.8·(0.200, 0.100, 0.500, 0.100, 0.200) = (0.160, 0.080, 0.400, 0.080, 0.160). Sum S = 0.880.

Teleport: $(1-\beta)/|S|$ = 0.2/2 = 0.10 to A and C; 0 to others. → $r^{(1)}$ = (0.260, 0.080, 0.500, 0.080, 0.160). Sum = 1.080 ≠ 1.

Why doesn't it sum to 1? Because the graph has no dead-ends in this case but the LEAK formula we just used is the simplified Topic-PR one. To preserve normalization with leaked mass S, use $(1-S)/|S|$ = 0.12/2 = 0.060 to seeds: $r^{(1)}$ = (0.220, 0.080, 0.460, 0.080, 0.160). Sum = 1.000 ✓.
Lesson: the simple $(1-\beta)/|S|$ only sums to 1 when $S = \beta \cdot M \cdot r^{(0)}$ sums to β. Otherwise use $(1-S)/|S|$.

A5

(a) $r^-_p = 0.025 - 0.020 = 0.005$. Spam Mass = 0.005/0.025 = **0.20**.
(b) 0.20 < 0.40 → **NOT flagged**. Most of p's rank comes from trusted sources.
(c) New $r^-_p = 0.025 - 0.022 = 0.003$. New Spam Mass = 0.003/0.025 = **0.12**. Spam Mass **decreased** by 0.20 − 0.12 = **0.08**. Larger trusted seed set means more of p's rank is now attributable to trusted sources.

A6

A = [[0,1,1],[1,0,1],[0,1,0]] (rows=source). A^T = [[0,1,0],[1,0,1],[1,1,0]]. $h^{(0)} = a^{(0)} = (1, 1, 1)$.

Iter 1. $a = A^T \cdot h = (h_2, h_1+h_3, h_1+h_2) = (1, 2, 2)$. L1 norm = 5. → a = (0.20, 0.40, 0.40). $h = A \cdot a = (a_2+a_3, a_1+a_3, a_2) = (0.80, 0.60, 0.40)$. L1 norm = 1.80. → h = (0.444, 0.333, 0.222).

Iter 2. $a = A^T \cdot h = (h_2, h_1+h_3, h_1+h_2) = (0.333, 0.667, 0.778)$. L1 = 1.778. → a = (0.187, 0.375, 0.438). $h = A \cdot a = (0.813, 0.625, 0.375)$. L1 = 1.813. → h = (0.448, 0.345, 0.207).

Strongest authority: page 3 (linked from both 1 and 2). **Strongest hub: page 1** (links to both 2 and 3 — most “index-page-like”).

A7

Implementation change: in step (c) of the W8-9 complete algorithm — “$r^{new}_j = r'^{new}_j + (1-S)/N$” — replace (1−S)/N with (1−S)/|S| if j ∈ S else 0.
Concrete example: graph 1→2, 2→1 (a 2-cycle) plus an isolated page 3 with no links. Vanilla PageRank gives r ≈ (0.4, 0.4, 0.2) — page 3 gets the teleport baseline. With S = {3}, all teleport mass goes to 3 and rank concentrates there: r ≈ (0.15, 0.15, 0.70). The two rankings disagree on the most important page.

A8

Trust attenuation — trust decreases with graph distance from the seed set. Algebraically: at every hop, only β fraction of trust is propagated; the remaining (1−β) is teleported. After k hops, only β^k of the original trust survives.

Trust splitting — a page p with k out-links gives only t_p/k trust to each. Algebraically: the $(1/d_j)$ factor in M_{ji}. Captures the editorial-rigor intuition: more out-links per page → less endorsement weight per link.

A9

Spam-farm derivation: target t's PageRank = x (from accessible) + $\beta \cdot M \cdot rank_{farm}$ (from M owned pages) + (1−β)/N (teleport). Each farm page receives β·y/M from t plus (1−β)/N teleport. Substituting: $y = x + \beta \cdot M \cdot (\beta y/M + (1-\beta)/N) + (1-\beta)/N$. The β^2 factor comes from t→farm (one β) and farm→t (another β). The (1−β)/N teleport is dropped because it's $O(1/N) \approx 10^{-9}$ — negligible compared to x and to βM/N. Multiplier for β = 0.9: 1/(1−0.81) = 1/0.19 ≈ **5.26**. Higher β → bigger multiplier → spam farms more effective.

A10

(a) Per-page output: PageRank → 1 score; TrustRank → 1 trust score; HITS → 2 scores (h, a); Spam Mass → 1 ratio in [0, 1].
(b) Teleport / normalization: PageRank → uniform teleport; TrustRank → teleport into trusted seed set; HITS → no teleport, explicit L1/L2 normalization; Spam Mass → derived from r and r^+, no iteration of its own.
(c) Question answered: PageRank — “how important is page p?” TrustRank — “how trusted?” HITS — “is p a hub or an authority?” Spam Mass — “what fraction of p's rank is spam?”

A11

Direct TrustRank thresholding misclassifies legitimate but isolated pages as spam — pages far from any seed naturally have low TrustRank regardless of legitimacy. Example: a legitimate small academic blog with low overall PageRank $r_p = 0.0001$ and TrustRank $r^+_p = 0.00009$. Direct TrustRank threshold of 0.001 flags this as spam. Spam Mass = (0.0001 − 0.00009)/0.0001 = 0.10 → not flagged (most of its small rank is from trusted sources, the page is just isolated, not spammy). Spam Mass normalizes by total rank, removing the “distance from seed” bias.

A12

(B) (1−β)/|S| — the teleport mass is divided among |S| seed pages.

A13

$1/(1 - 0.5^2) = 1/0.75 = 1.333$. **(B)**.

A14

False. Without normalization, h and a grow without bound (or shrink to zero). The eigenvalue of $A \cdot A^T$ is generally not 1 — it could be > 1 (so values explode) or < 1 (values vanish). Explicit L1 or L2 normalization at each step keeps the iteration bounded and convergent to the principal eigenvector direction.

A15

(D). Controlled-membership domains (.edu, .gov, .mil) provide high-trust seeds with negligible spam. (B) is also reasonable but top-PR pages can include compromised popular sites. (A) is wasteful (seeds may be mostly low-trust). (C) is impossible — humans can't vet billions of pages.

§ 12 ENDING KEY NOTES — REVISION CARDS

PageRank's 3 weaknesses Topic blindness, single importance score, link-spam vulnerability.	**Topic-Sensitive PR** $r = \beta \cdot M \cdot r + (1-\beta) \cdot v_S$, $v_S[j]=1/\lvert S\rvert$ if $j \in S$ else 0.
Topic-PR teleport $(1-\beta)/\lvert S\rvert$ added to every $j \in S$; 0 to other pages.	**Algorithm change** ONE line: redistribute leaked/teleport mass to S only, not all N pages.
Term spam Hidden keyword stuffing, copying top results invisibly.	**Google's term-spam fix** Trust anchor text from incoming links, not page's own body text.
3 page types (spammer view) Inaccessible / Accessible / Owned.	**Spam farm goal** Maximize PageRank of target page t by mutual links with M owned pages.
Spam farm formula $y = x/(1-\beta^2) + c \cdot M/N$, $c = \beta/(1+\beta)$.	**Multiplier (β=0.85)** $1/(1-\beta^2) \approx 3.6$ — small accessible-PR x amplified ~3.6×.
TrustRank Topic-sensitive PR with teleport set = trusted seed pages.	**Trust attenuation** Trust decays β per hop; far-from-seed pages get little trust.
Trust splitting Trust divided across out-links: page with k out-links → t/k each.	**Seed set choices** (1) top-k by PageRank, (2) controlled domains (.edu/.gov/.mil).
Spam Mass formula $SM(p) = (r_p - r^+_p)/r_p$.	**Spam Mass intuition** Fraction of p's rank coming from non-trusted (likely spam) sources.
HITS scores h_i = hub score, a_j = authority score.	**HITS iteration** $h \leftarrow A \cdot a$, $a \leftarrow A^T \cdot h$. Normalize each step.
HITS eigen-equation $a = A^T A \cdot a$ (principal eigenvector); $h = AA^T \cdot h$.	**PageRank vs HITS** PR: 1 score, query-independent, teleport. HITS: 2 scores, query-specific, normalized.

§ 13 FORMULA & ALGORITHM REFERENCE

Quick-look reference for everything advanced in W12.

Concept	Formula / Statement	Use
Topic-Sensitive PR	$r = \beta \cdot M \cdot r + (1-\beta) \cdot v_S$, $v_S[j]=1/\lvert S\rvert$ if $j \in S$	Bias rank to a specific topic.
Topic-PR teleport	$(1-\beta)/\lvert S\rvert$ to seeds only, 0 to others	One-line modification of W8-9 algorithm.
Spam farm rank	$y = x/(1-\beta^2) + (\beta/(1+\beta)) \cdot M/N$	Estimating effectiveness of a link farm.
Multiplier (β=0.85)	$1/(1-\beta^2) \approx 3.6$	How much accessible PR is amplified.
TrustRank	Same as Topic-PR with seed set = trusted pages	Demoting link-spam farms automatically.
Trust propagation	trust q = $\beta \cdot \Sigma_{p \to q}$ trust$_p/\lvert O_p\rvert$ + teleport	Per-iteration update.
Spam Mass	$SM(p) = (r_p - r^+_p)/r_p$	Fractional spam contribution; threshold to label spam.
HITS authority	$a = A^T \cdot h$; converges to principal eigenvector of A^TA	Pages that are linked-to by good hubs.
HITS hub	$h = A \cdot a$; converges to principal eigenvector of AA^T	Pages that link-to good authorities.

Table 13.1. Advanced link-analysis reference.

Connections to other weeks:

- **W8-9 PageRank**: every algorithm here is a parametrization of the W8-9 power iteration. Topic-Sensitive PR changes the teleport vector. TrustRank inherits Topic-PR. Spam Mass is post-hoc.
- **W16 Community Detection**: spectral clustering uses the same eigenvector machinery on a different matrix (the graph Laplacian instead of M). HITS's A^TA is the link-graph analog.
- **MMDS §5.3–5.5**: the textbook's depth on Topic-PR, Link Spam, TrustRank, Spam Mass, and HITS. Read for exercise variety, not for new concepts beyond what's here.